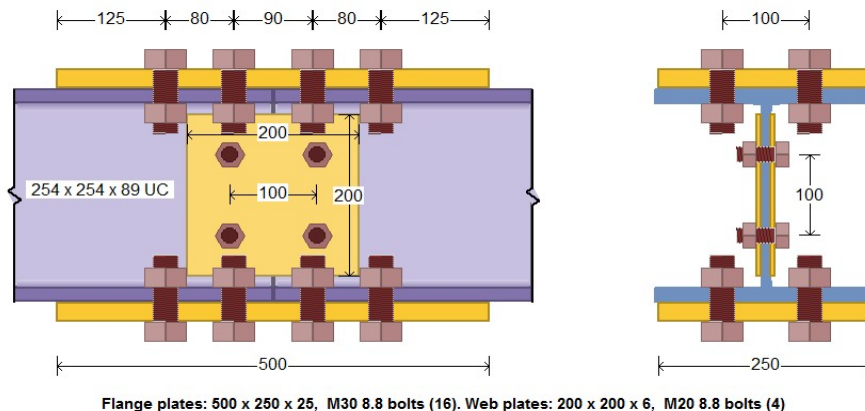


Flange plate splice calculations

Beam: Upper Ground Spine

Beam span: 6.6 m. Section: **254 x 254 x 89 UC** S355



Splice 1: 2.4 m. from R1. BM: 185.7 kNm SF: 39.93 kN (factored)

Splice 2: 4.4 m. from R1. BM: 200.0 kNm SF: 26.37 kN (do.)

Section size: **254 x 254 x 89 UC S355**

Section dims: D = 260.3 B = 256.3 T = 17.3 t = 10.3 d = 200.3 r = 12.7

Use **S275** flange plates **500 x 250 x 25 mm**,

Use 2 pairs of M30 8.8 bolts at 100 mm cross centres; inmost pairs of bolts 90 mm apart, 1 further sets of bolts at 80 mm pitch

Web plates: 200 x 200 x 6 mm, 2 pairs of M20 8.8 bolts at 100 mm (H) and 100 mm (V)

Stiffness check: *SCI AD243 recommends that flange plate I values should be not less than beam values*

Beam/plate I_x : 14,300/25,501 cm⁴; I_y : 4,860/6,510 cm⁴

Basic detailing checks

Flange/Flange Plates: Using M30 bolts, washer diameter 56 mm

Check flange bolt washer clear beam web: pitch (100 mm) $\geq$ 92 mm OK

Flange plate bolt edge distance (75.0 mm) $\geq$ 1.25D (37.5 mm) OK

Flange bolt edge distance (78.1 mm) $\geq$ 1.25D (37.5 mm) OK

Plate plate bolt end distance (125 mm) $\geq$ 1.25D (37.5 mm) OK

Flange bolt end distance (45 mm) $\geq$ 1.25D (37.5 mm) OK (n.b. bearing capacity is reduced as $< 2.0d$)

Flange bolt spacing (80 mm) $\geq$ 2.5D (75 mm) OK

Web/Web Plates: Using M20 bolts, washer diameter 37 mm

Check web plate clears beam roots: plate height, 200 mm $<$ d, 200.3 mm OK

Web plate bolt end distance H (50 mm) $\geq$ 2.0D (40 mm) OK

Web plate bolt end distance V (50 mm) $\geq$ 2.0D (40 mm) OK

Web plate bolt pitch (100 mm) $\geq$ 2.5D (50 mm) OK

Web bolt end clearance (50 mm) $\geq$ 2.0D (40 mm) OK

All dimensional checks satisfied

Flange plates

Check splice with BM = 200.0 kNm

Axial force in beam flanges = $200.0 \times 1000 / (260.3 - 17.3) = 823$ kN

Beam flange area, gross = 4,434 mm², net = 3,327 mm²; Flange plate area: gross = 6,250 mm², net = 4,650 mm²

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Axial capacities: Plate: $6,250 \times 265/1000 = 1656 \text{ kN}$
 Plate net: $1.2 \times 4,650 \times 265/1000 = 1479 \text{ kN}$
 Beam flange: $4,434 \times 345/1000 = 1530 \text{ kN}$
 Beam flange net: $1.1 \times 3,327 \times 345/1000 = 1263 \text{ kN} <<< \text{OK}$

Plate block shear: $0.6 \times 265 \times 25 \times [205 + 1.2 \times (175 - 2.5 \times 32)]/1000 = 1,268 \text{ kN OK}$ ($K_e = 1.2$; $k = 2.5$) [6.2.4]

Strut action moment: Flange effective area: $3,659 \text{ mm}^2$ Plate effective area: $5,580 \text{ mm}^2$
 Flange modulus, $Z_y: 152.5 \text{ cm}^3$ Plate modulus, $Z_y: 207.1 \text{ cm}^3$
 Check for strut action moment not required

Bolts: M30 8.8, net CSA 561 mm^2 , $p_s = 375 \text{ N/mm}^2$, $p_{bs} = 460/550 \text{ N/mm}^2$ (S275/S355) [Tables 30/32]
 Shear resistance per bolt: $561 \times 375/1000 = 210 \text{ kN}$
 Beam flange bearing resistance per bolt: $30 \times 17.3 \times 550/1000 = 285 \text{ kN}$
 Flange end bolt capacity (reduced as $e < 2d$) = $0.5 \times 45 \times 17.3 \times 550/1000 = 214 \text{ kN}$ [6.3.3.3]
 Beam flange bolt group resistance = $2 \times 2 \times 210 = 842 \text{ kN}$ (shear governs)
 Flange plate bearing resistance per bolt: $30 \times 25 \times 460/1000 = 345 \text{ kN}$
 Flange plate bolt group resistance = $2 \times 2 \times 210 = 842 \text{ kN}$ (shear governs)
 Moment capacity = $842 \times 243.0/1000 = 204.5 \text{ kNm}$ (limited by bolt group (flange))
Splice moment capacity = 204.5 kNm OK

Web plates

Check splice with SF = 39.93 kN

Shear: Gross area, $A_v = 2 \times 0.9 \times 200 \times 6 = 2,160 \text{ mm}^2$ [4.2.3]
 (2 plates) Net area, $A_{v,net} = 2,160 - (2 \times 2 \times 22 \times 6) = 1,632 \text{ mm}^2$
 As $A_{v,net} \geq 0.85 A_v / K_e$ ($1,530$, $K_e = 1.2$), gross area, A_v can be used [6.2.3]
 Shear capacity = $0.6 \cdot p_y \cdot A_v = 0.6 \times 2,160 \times 275/1000 = 356 \text{ kN OK}$ [4.2.3]

Plate block shear: $2 \times 0.6 \times 265 \times 6 \times [101 + 1.2 \times (100.0 - 0.5 \times 32)]/1000 = 396 \text{ kN OK}$ ($K_e = 1.2$; $k = 0.5$)

Bolts: M20 8.8, net CSA 245 mm^2 , $p_s = 375 \text{ N/mm}^2$, $p_{bs} = 460/550 \text{ N/mm}^2$ (S275/S355) [Tables 30/32]
 Shear capacity per bolt = $2 \times 245 \times 375/1000 = 184 \text{ kN}$ (double shear)
 Bearing on web plates per bolt: $2 \times 20 \times 6 \times 460/1000 = 110 \text{ kN} <<<$
 Bearing on beam web per bolt: $20 \times 10.3 \times 550/1000 = 113 \text{ kN}$

Load per bolt Shear on each bolt = $39.93/2 = 19.97 \text{ kN}$
 Modulus of web bolt group = $n(n+1)p/6 = 2 \times 3 \times 100/6 = 100 \text{ mm}$
 Moment load on outermost bolts = $39.93 \times 50/100 = 19.97 \text{ kN}$
 Load on outermost bolts = $\sqrt{(19.97^2 + 19.97^2)} = 28.24 \text{ kN OK}$
 Web plate bolts are adequate
Splice shear capacity = 156.1 kN OK